

GHRCEM WAGHOLI PUNE
DEPARTMENT OF AI & AIML
SY CSE AI,AIML,CS & DS
QUESTION BANK FOR CAE I
SUBJECT: DMGT (UBSL205)

Sr no	Question	Unit	BL	CO	PO
1	Let $R = \{(1,4), (2,1), (2,5), (2,4), (4,3), (5,3), (3,2)\}$ on the set $= \{1,2,3,4,5\}$. Use the WARSHALL's Algorithm to find the Transitive closure of R	I	3	1	1
2	Out of 80 players 35 play cricket,42 play hockey,25 play football,8 play both cricket and hockey,10 play both cricket and football, 7 play both hockey and football,3 play all 3 games. (i) how many players are playing only one game? (ii) how many players are playing exactly two games? (iii) how many players are playing exactly three games?	I	5	1	1
3	Use Mathematical induction to show that 2 divides $n^2 + n$ whenever n is positive integer	I	4	1	1
4	If $A = \{1,2,3,4\}$, $R = \{(1,2), (2,1), (2,3), (3,4), (4,1)\}$ Using Warshall's Algorithm Find the transitive closure of R	I	3	1	1
5	A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, one in Java and Python, five in C# and Python, and just one programmer is proficient in all three languages above.Determine the number of computer programmers that are not proficient in any of these three languages.	I	5	1	2
6	Draw the Hasse diagram representing the partial ordering $\{(a,b): a \text{ divides } b\}$ on $\{1,2,3,4,6,8,12\}$	I	4	1	1
7	If $A=\{1,2,3,4\}$, $R=\{(2,1), (2,3), (3,1), (3,4),(4,1),(4,3)\}$ Using Warshall Algorithm Find the transitive closure of R	I	3	1	1
8	Draw the Hasse diagram for the less than or equal to relation on $\{0, 2, 5, 10, 11, 15\}$	I	4	1	1
9	Draw the Hasse diagram for the Poset $(p(S), \subseteq)$, where $p(S)$ is the	I	4	1	1

	power set of set $S = \{a, b, c\}$				
10	If $A = \{1,2,3,4\}$, $R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$ Using Warshall's Algorithm Find the transitive closure of R .	I	3	1	1
11	If $A = \{1,2,3,4\}$, $R = \{(1,1), (2,2), (3,3), (3,4), (4,4), (4,3)\}$ then show that R is an equivalence relation on set A	I	3	1	2
12	Prove that, by using the principle of mathematical induction, $2n > n$ for all positive integers n	I	4	2	1
13	Show that Group $G = \{1, -1, i, -i\}$ is cyclic Group under multiplication. State the generates of G,	II	5	2	1
14	If $G = (Z_4, +_4)$ is Group and $H = \{0,2\}$ is subgroup of G , then Find all cosets of H in G , Is H is Normal Subgroup of G ?	II	5	2	1
15	Using Fermat's theorem find the remainder when 7^{222} is divide by 11	II	3	2	1
16	Show that Group $G = \{1, -1, i, -i\}$ is cyclic Group under multiplication. State the generates of G, Is $H = \{1, -1, i\}$ is Subgroup of G under multiplication ? Justify your answer.	II	5	2	1
17	If $G = (Z_6, +_6)$ is Group and $H = \{0,3\}$ is subgroup of G , then Find all cosets of H in G , Is H is Normal Subgroup of G ? If Yes then find Quotient group G/H	II	5	2	2
18	Consider $Z_5 = \{0,1,2,3,4\}$ i) Show that $(Z_5, +_5)$ is Group ii) Is $(Z_5, +_5)$ is abelian Group ? Justify your answer iii) Is $H = \{0,1\}$ subgroup of Z_5 under the binary operation addition modulo 5?	II	5	2	1
19	Let G be the set of all positive rational numbers and * be the binary operation on G defined by $a * b = \frac{ab}{7}$ for all $a, b \in G$. Prove that $(G, *)$ is abelian Group ?	II	3	2	2
20	Using Fermat's theorem find the remainder when 7^{2001} is divide by 5	II	3	2	1

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SUBJECT INCHARGE

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